

Multiple Choice (10 marks)

1. Calculate the Laplace transform of $t^n u(t)$.

- (a) $L\{t^n u(t)\} = [n! / (s^{n+3})]$
- (b) $L\{t^n u(t)\} = [(n+1)! / (s^{n+1})]$
- (c) $L\{t^n u(t)\} = [n! / (s^n)]$
- (d) $L\{t^n u(t)\} = [n! / (s^{n+2})]$
- (e) $L\{t^n u(t)\} = [n! / (s^{n+1})]$

2. A response transform $V_{ab}(s)$ is related to a source transform $V(s)$ by $V_{ab}(s) = [3V(s) + 3s + 27] / (s^2 + 6s + 8)$. Find $v_{ab}(t)$ if $v(t) = 16e^{-3t}u(t)$.

- (a) $-48e^{-3t} - (68/2)e^{-2t} + (33/2)e^{-4t}$
- (b) $16e^{-3t} + (69/2)e^{-2t} + (33/2)e^{-4t}$
- (c) $-48e^{-3t} + (69/2)e^{-2t} + (33/3)e^{-4t}$
- (d) $-48e^{-3t} + (70/2)e^{-2t} + (33/2)e^{-4t}$
- (e) $-49e^{-3t} + (69/2)e^{-2t} + (33/2)e^{-4t}$

3. Find the attenuation per meter along a wave guide for an applied wave length $\lambda_0 = 2$ meters, if the cutoff wave length of the guide is $\lambda_{oc} = 20$ cm.

- (a) 400 db/meter
- (b) 545 db/meter
- (c) 600 db/meter
- (d) 325 db/meter
- (e) 250 db/meter

4. Compute the impact stress in a steel rod which is hit by a block weighing 45 lbs. The speed of the block is 40 in./sec. The rod is 18 in. long and has a diameter of $1(1/4)$ in.

- (a) 8,762.3 psi
- (b) 4,321.0 psi
- (c) 25,678.9 psi
- (d) 6.25 lbs
- (e) 30,000 psi

5. In decimal number system what is MSD

- (a) First two digits from right to left
- (b) Median of all digits
- (c) First digit from right to left
- (d) First digit from left to right
- (e) Mode of all digits

True / False (10 marks)

Answer True or False

- 6. True or False: Silicon is regarded as a highly detrimental impurity in magnetic materials and signal processing.
- 7. True or False: The theoretical variance of the given probability distribution is 1.125.
- 8. True or False: To drive a load at 1750 r/min with a torque of 60 lb.ft, a horsepower of 20 hp is required.
- 9. True or False: A single phase full bridge inverter can operate in load commutation mode when the load is inductive.
- 10. True or False: The power delivered by a roller chain of 3/4 in. pitch on a 34-tooth sprocket at 500 rpm is 1.5 hp.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss the significance of temperature measurement in a chemistry laboratory, specifically conversion of units and its implications for experimental accuracy.
- 12. Given the current $i_1(t) = 2e^{t/10}$ A flowing into the positive terminal of a circuit element, analyze the power dissipation in the element over the first 10s using the provided voltage equations. Explain your reasoning and calculations in detail.

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